

TrES-3b

R-0481

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_180818 · created 2026-07-20T06:44:50+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458348.7248 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.047
Transit depth (Rp/Rs)^2	3.86 +/- 1.84 [%]
Orbital Inclination (inc)	86.99 +/- 2.3
Airmass coefficient 1 (a1)	0.99 +/- 0.11
Airmass coefficient 2 (a2)	0.015 +/- 0.083
Scatter in the residuals of the lightcurve fit is	10.55 %
Best Comparison Star	#2 - [271, 354]
Optimal Aperture	5.64
Optimal Annulus	9.85
Transit Duration (day)	0.0814 +/- 0.0074

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.196 $\pm$ 0.047 vs 0.16309 (deviation 0.51 σ)
Duration fit vs published	0.0814 d vs 0.0567 d (deviation 2.43 σ)
Mid-time O-C	-3.7 min
Depth SNR	3.0
Residual scatter	10.55 %

Light curve

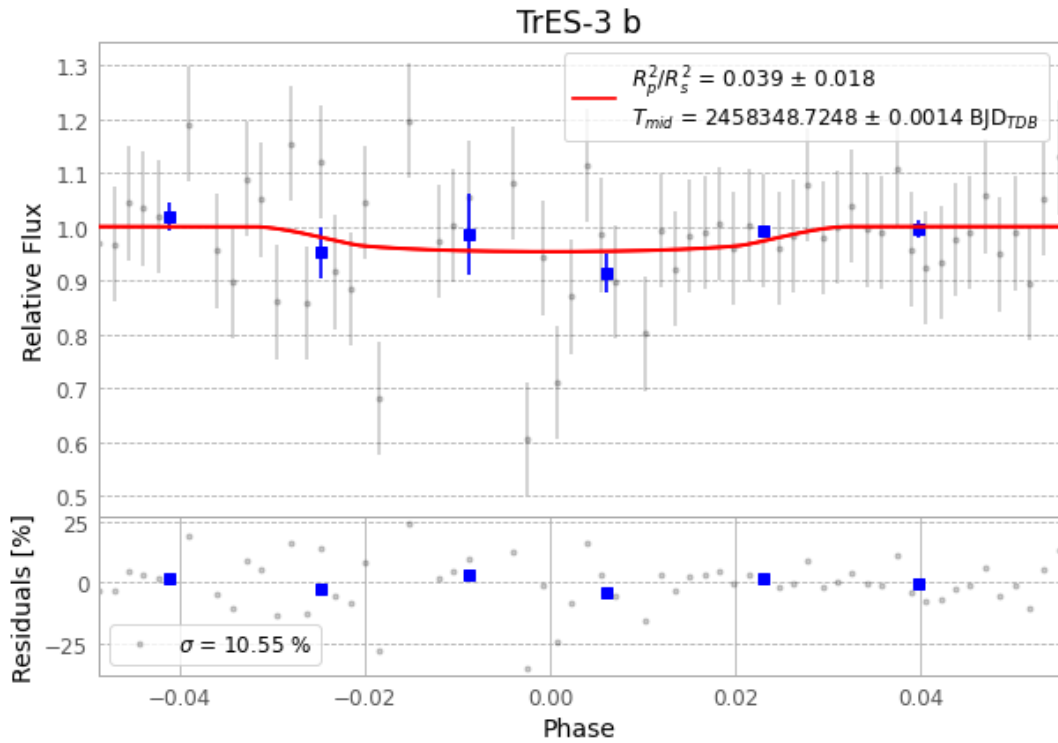


Figure 1 — FinalLightCurve_TrES-3 b_2018-08-17.png (SHA-256 85b18e950da5ed59...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [341, 269], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 08cf456489311d5e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

64 FITS frames (42 MB), TRES-3180818034812.FITS → TRES-3180818070312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-180818.log (1424118d322b...), FinalLightCurve_TrES-3 b_2018-08-17.png (85b18e950da5...), FinalLightCurve_TrES-3 b_2018-08-17.csv (655593de4b74...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 06:44Z.